MNIST Classification: MLE and Discriminant Analysis

1. Data Preprocessing and Sampling

The MNIST dataset was filtered to include only classes 0, 1, and 2. From each class, 100 random samples were selected for the training set (300 total) and 100 for the test set (300 total). Images were flattened into 784-dimensional feature vectors and normalized to the range [0, 1].

2. MLE Estimation and Classification Accuracy

Mean and covariance matrices were estimated using Maximum Likelihood Estimation (MLE) formulas as specified in the lecture handouts. Classification was performed using Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA).

Classifier	**Accuracy (%)**
LDA	91.67%
QDA	98.00%

3. Discriminant Values for Sample Test Point 0

The log-likelihood discriminant values for Sample 0 (True Label: 0) were calculated manually. Class 0 yielded the highest value, leading to a correct prediction.

Class 0: -2,361,810.6431
Class 1: -2,418,952.9385
Class 2: -10,456,013.7369

4. t-SNE Visualizations

Figure 1: t-SNE Plot of the MNIST Training Set (Digits 0, 1, 2)

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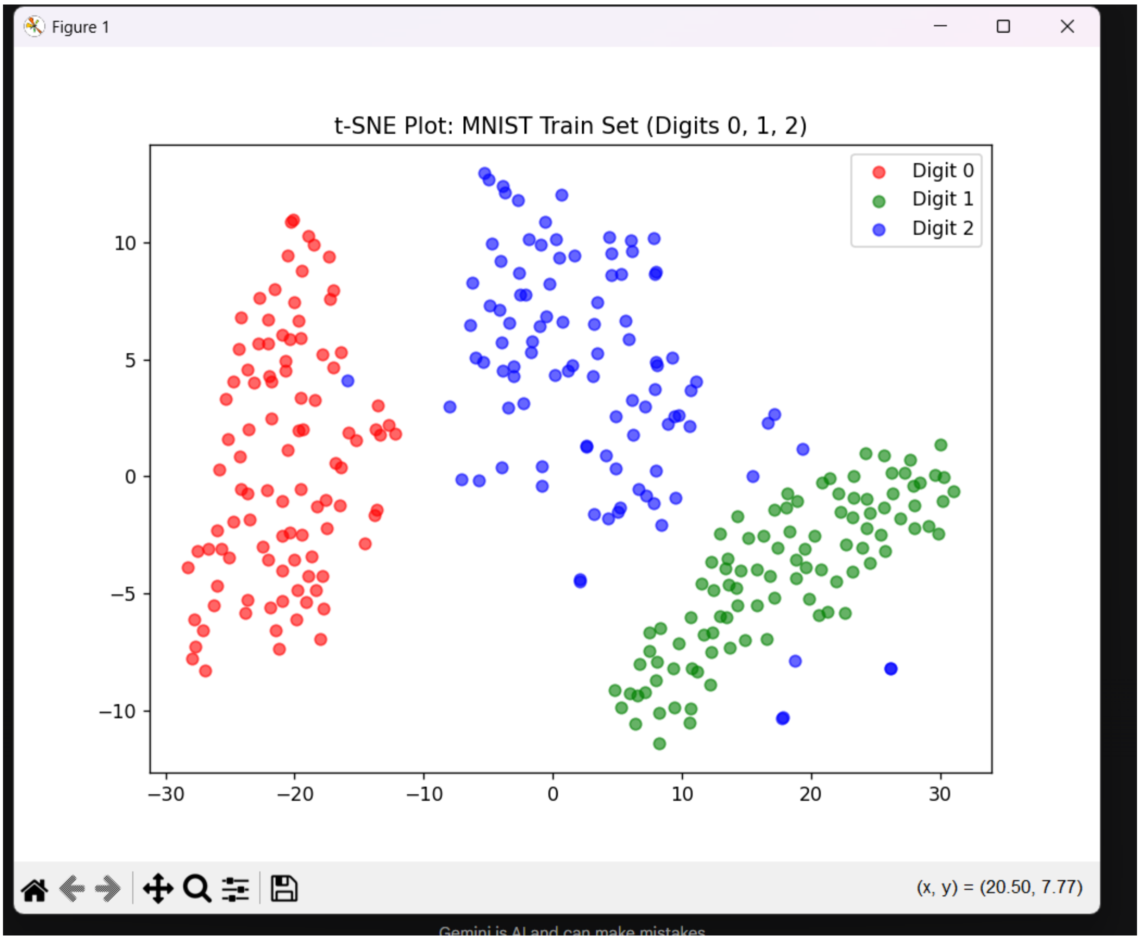

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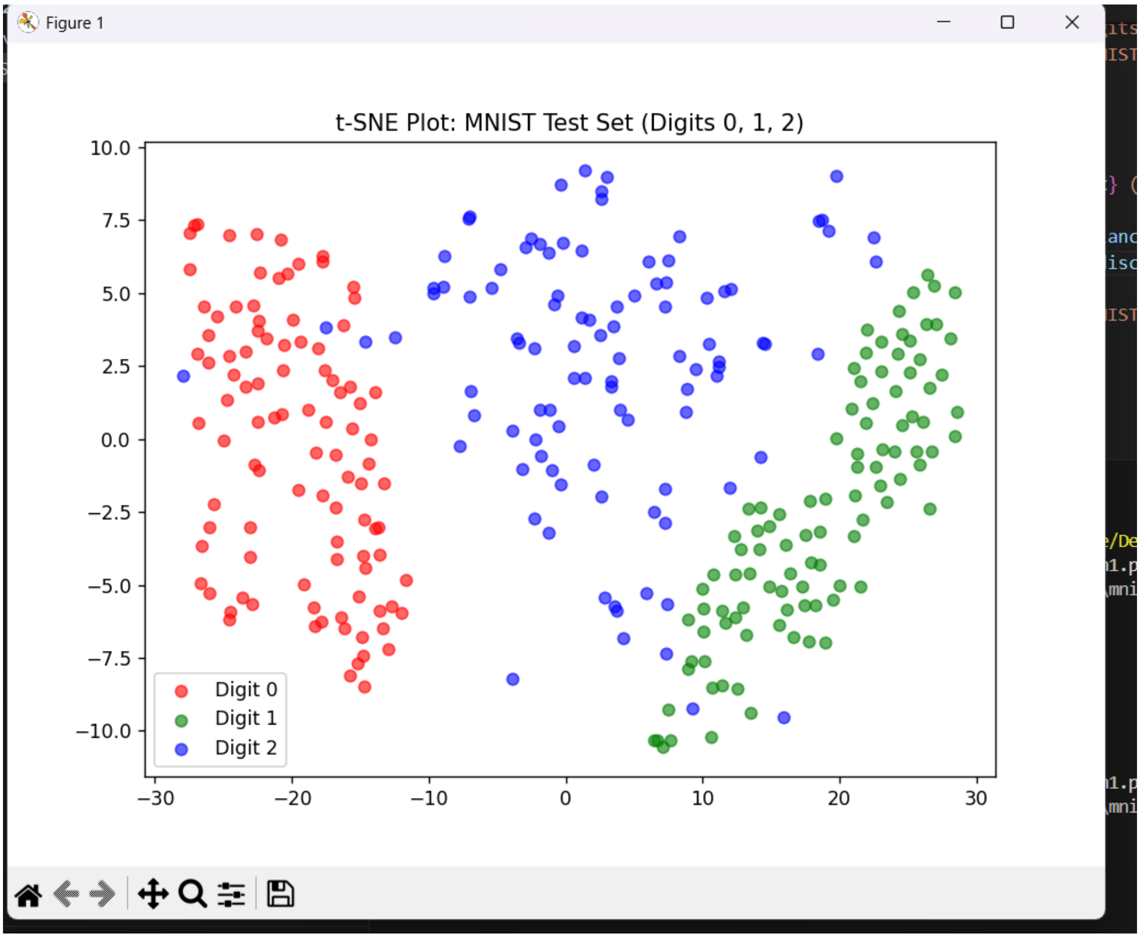
Figure 2: t-SNE Plot of the MNIST Test Set (Digits 0, 1, 2)



5. Conclusion

QDA outperformed LDA (98.00% vs 91.67%) because it does not assume a shared covariance matrix. This allows the model to better capture the specific pixel distribution patterns unique to each digit.